particular, for any cofibered groupoid A *over* $\underline{C}_\Lambda$ *, a normalized prorepresentation, if it exists, is unique up to an isomorphism.*

Proof: Consider the commutative diagram

$$\begin{tikzcd} \underline{h}_{(R_0,R_1,\dots)} \arrow[rr, "\underline{h}_{(\alpha_0,\alpha_1)}"] & & \underline{h}_{(R_0,R_1,\dots)} \\ \\ \underline{h}_{R_0} \arrow[uu, "\varepsilon"] \arrow[rr, "\underline{h}_{\alpha_0}"'] & & \underline{h}_{R_0} \arrow[uu, "\varepsilon"'] \end{tikzcd}$$

where $\underline{h}_{(\alpha_0,\alpha_1)}$ is a $\underline{C}_\Lambda$-equivalence. Since $\underline{h}_{(R_0,R_1,\dots)}(k[\varepsilon])$ is totally disconnected, $[\underline{h}_{(\alpha_0,\alpha_1)}(k[\varepsilon])$ and $[\varepsilon(k[\varepsilon])]$ are both bijections, and hence so is $\underline{h}_{\alpha_0}(k[\varepsilon])$.

Consequently, $\alpha_0 : R_0 \to R_0$ is a surjective endomorphism and hence is an automorphism. It then follows that α_1 is also an automorphism of R_1 .

Lemma 2.15. *Let* A, B *be discrete groupoids over* $\underline{C}_\Lambda$ *and* $\varphi : A \to B$ *a minimally smooth functor. If* A, B *are homogeneous, then* φ *is a* $\underline{C}_\Lambda$*-equivalence.*

Proof: We must show that $\varphi(S) : A(S) \longrightarrow B(S)$ is bijective for all $S \in \mathrm{ob}(\underline{C}_\Lambda)$. We note that $\varphi(S)$ is surjective for all $S \in \mathrm{ob}(\underline{C}_\Lambda)$ and bijective for $S = k[\varepsilon]$. We argue by induction on the length of S . Let $S' \xrightarrow{f} S$ be a small extension in $\underline{C}_\Lambda$. Since A, B are homogeneous discrete groupoids, the canonical isomorphism $1 \times \tilde{f} : S' \underset{S}{\times} S' \longrightarrow S' \underset{k}{\times} k[\mathrm{Ker}\ f]$ induces the commutative diagram

$$\begin{array}{ccc} A(S') \times_{A(S)} A(S') & \xrightarrow{\simeq} & A(S') \times A(k[\mathrm{Ker}\ f]) \\ \downarrow & & \downarrow \\ B(S') \times_{B(S)} B(S') & \xrightarrow{\simeq} & B(S') \times B(k[\mathrm{Ker}\ f]) \end{array}$$

Therefore $A(S')$, $B(S')$ are principal homogeneous sets over $A(S)$, $B(S)$ with the structural group $A(k[\mathrm{Ker}\ f]) \xrightarrow{\simeq} B(k[\mathrm{Ker}\ f])$. Consequently if $\varphi(S)$ is bijective, then $\varphi(S')$ is injective and hence is bijective.

Corollary 2.16. Let $\underline{A}$ be a homogeneous groupoid over $\underline{C}_\Lambda$ such that $\mathrm{Aut}_{\underline{A}(k[\varepsilon])}(1_\varepsilon)$ is of finite-dimensional. Then for any $\xi \in ob(\hat{\underline{A}})$, the functor $\mathrm{Isom}_\xi : \underline{C}_{R\hat{\otimes}_\Lambda R} \longrightarrow (\mathrm{Ens})$ is prorepresentable, where $R = P(\xi)$.

Proof: Assume that Isom_ξ is homogeneous. We then have, by 1.11, a minimally smooth functor $\varnothing : \underline{h}_{R_1} \to \underline{\mathrm{Isom}}_\xi$, which is a $\underline{C}_{R\hat{\otimes}R}$-equivalence by 2.15 i.e., Isom_ξ is prorepresentable. Therefore it suffices to show that Isom_ξ is homogeneous. Consider a diagram

$$\begin{array}{ccc} T_1 & & T_2 \\ & \searrow \quad \swarrow & \\ & T & \end{array}$$

in $\underline{C}_{R\hat{\otimes}_\Lambda R}$ where $T_2 \longrightarrow T$ is surjective. Since $\underline{A}$ is homogeneous, $\xi_i(T_1) \times_{\xi_i(T)} \xi_i(T_2)$ exists in $\underline{A}$, and we have a canonical isomorphism

$\xi_i(T_1 \times_T T_2) \simeq \xi_i(T_1) \times_{\xi_i(T)} \xi_i(T_2)$ for $i = 1,2$ (cf. 2.6(a)). It follows that

$\mathrm{Isom}_\xi(T_1 \times_T T_2) \longrightarrow \mathrm{Isom}_\xi(T_1) \times_{\mathrm{Isom}_\xi(T)} \mathrm{Isom}_\xi(T_2)$ is an isomorphism.

Theorem 2.17. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$. Then it is prorepresentable by a (normalized) smooth cogroupoid if and only if

(1) $\underline{A}$ is homogeneous

(2) $[A(k[\varepsilon])]$ and $[\mathrm{Aut}(1_\varepsilon)]$ are both finite-dimensional vector spaces over k .

Proof: Assume that $\underline{A}$ is prorepresentable by a smooth cogroupoid i.e., $\underline{A}$ is $\underline{C}_\Lambda$-equivalent to $\underline{h} = \underline{h}_{(R_0,R_1,\dots)}$ for some smooth cogroupoid $(R_0,R_1,\dots)$ in $\hat{\underline{C}}_\Lambda$. Since $\underline{h}$ is homogeneous by 2.12(a), and

$$[\underline{h}(k[\varepsilon])] = \mathrm{Coker}\ (\mathrm{Hom}_{\Lambda\text{-alg}}(R_1,k[\varepsilon] \overset{s_1}{\underset{s_2}{\rightrightarrows}} \mathrm{Hom}_{\Lambda\text{-alg}}(R_0,k[\varepsilon]))$$ as well as

$$\begin{aligned}\mathrm{Aut}_{\underline{h}}(1_\varepsilon) &= \mathrm{Ker}(\mathrm{Hom}_{\Lambda\text{-alg}}(R_1,k[\varepsilon]) \xrightarrow{s_1} \mathrm{Hom}_{\Lambda\text{-alg}}(R_0,k[\varepsilon]))\\ &= \mathrm{Ker}(\mathrm{Hom}_{\Lambda\text{-alg}}(R_1,k[\varepsilon]) \xrightarrow{s_2} \mathrm{Hom}_{\Lambda\text{-alg}}(R_0,k[\varepsilon]))\end{aligned}$$

are both finite-dimensional vector spaces over k , we must have the same properties on $\underline{A}$. Conversely assume (1) and (2) on $\underline{A}$. We claim that $\underline{A}$ is prorepresentable by a normalized smooth cogroupoid. Since $\underline{A}$ is homogeneous and $[\underline{A}(k[\varepsilon])]$ is a finite-dimensional vector space over k , $\underline{A}$ admits a minimally versal formal object ξ . Set $P(\xi) = R_0$ and consider the functor $\mathrm{Isom}_\xi : \underline{C}_{R_0 \hat{\otimes}_\Lambda R_0} \to (\mathrm{Ens})$ given by

$$T \longmapsto \mathrm{Isom}_{\underline{A}(T)}(\xi_1(T),\xi_2(T))$$

where $\xi_i(T) = \xi(f_i)$ and $f_i : R_0 \to T$ denotes the composit maps

$$R_0 \overset{s_1}{\underset{s_2}{\rightrightarrows}} R_0 \hat{\otimes}_\Lambda R_0 \dashrightarrow T \ .$$

This functor is pro-representable by 2.16, i.e. there exists $R_1 \in \mathrm{ob}(\widehat{\underline{C}_{R_0 \hat{\otimes}_\Lambda R_0}})$ and an isomorphism $\xi_1(R_1) \xrightarrow{\varphi} \xi_2(R_1)$ such that $\varnothing : h_{R_1} \to \mathrm{Isom}_\xi$ is an isomorphism over $\underline{C}_{R_0 \hat{\otimes}_\Lambda R_0}$. Then $(R_0,R_1,\dots)$ defines a cogroupoid in $\hat{\underline{C}}_\Lambda$ and the functor $(\xi,\varphi) : \underline{h}_{(R_0,R_1,\dots)} \to \underline{A}$ is a $\underline{C}_\Lambda$-equivalence. Since ξ is a minimally versal formal object of $\underline{A}$, the composit functor $\underline{h}_{(R_0,R_0,\dots)} \xrightarrow{\varepsilon} \underline{h}_{(R_0,R_1,\dots)} \xrightarrow{(\xi,\varphi)} \underline{A}$ is, by definition, minimally smooth and therefore so is ε since (ξ,φ) is a $\underline{C}_\Lambda$-equivalence i.e., $(R_0,R_1,\dots)$ is a normalized smooth cogroupoid in $\hat{\underline{C}}_\Lambda$.

Corollary 2.18. *A functor* $F : \underline{C}_\Lambda \longrightarrow (\mathrm{Ens})$ *is pro-representable if and only if it is homogeneous and* $\dim F(k[\epsilon]) < \infty$.

Corollary 2.19. *Every smooth cogroupoid in* $\hat{\underline{C}}_\Lambda$ *can be normalized up to isomorphism.*

Given a cogroupoid $(R_0, R_1, \ldots)$ in $\hat{\underline{C}}_\Lambda$, we set $\bar{R}_1 = R_1/J$ where J is the ideal in R_1 generated by $\{s_1(r) - s_2(r) \mid r \in R_0\}$. Then $(R_0, \bar{R}_1, \ldots)$ defines a formal group scheme over R_0 with respect to the induced composition rule. Now let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$ prorepresented by a normalized smooth cogroupoid $(R_0, R_1, \ldots)$ via (ξ, φ) . Let $\underline{C}_k \hookrightarrow \underline{C}_{R_0} \hookrightarrow \underline{C}_{R_0 \hat{\otimes}_\Lambda R_0}$ be the embeddings corresponding to the surjective maps $R_0 \hat{\otimes}_\Lambda R_0 \longrightarrow R_0 \longrightarrow k$, and let Aut_ξ , Aut^0 be the restriction functors of Isom_ξ on $\underline{C}_{R_0}$, $\underline{C}_k$ respectively. In other words, $\mathrm{Aut}_\xi(S)$ = the automorphism group of $\xi(S)$ for every $S \in \mathrm{ob}(\underline{C}_{R_0})$, and $\mathrm{Aut}^0(S)$ = the automorphism group of 1_S for every $S \in \mathrm{ob}(\underline{C}_k)$, where 1_S = the direct image of $1 \in \underline{A}(k)$ under the map $k \to S$. Since R_1 prorepresents the functor Isom_ξ on $\underline{C}_{R_0 \hat{\otimes}_\Lambda R_0}$, it follows that $\bar{R}_1$ prorepresents the functor Aut_ξ on $\underline{C}_{R_0}$, and $k \hat{\otimes}_{R_0} \bar{R}_1$ prorepresents Aut^0 on $\underline{C}_k$. Though $\underline{A}$ is prorepresented by a smooth cogroupoid $(R_0, R_1, \ldots)$, neither $(R_0, \bar{R}_1, \ldots)$ nor $(k, k \otimes_{R_0} \bar{R}_1, \ldots)$ are in general smooth. Indeed we have

Corollary 2.20. *Let* $\underline{A}$ *be a homogeneous cogroupoid over* $\underline{C}_\Lambda$ *prorepresented by a normalized smooth cogroupoid* $(R_0, R_1, \ldots)$ *via* (ξ, φ) .

(a) $(R_0, \bar{R}_1, \ldots)$ *and* $(k, k \otimes_{R_0} \bar{R}_1, \ldots)$ *prorepresent the functors* Aut_ξ *and* Aut^0 *respectively.*

(b) *The following statements are equivalent*

(i) *the formal group scheme* $\underline{h}_{(R_0, \bar{R}_1, \ldots)}$ *is formally smooth over* R_0 ;

(ii) *for every surjective arrow* $a' \to a$ *in* $\underline{A}$,

$\mathrm{Aut}_{P(a')}(a') \longrightarrow \mathrm{Aut}_{P(a)}(a)$ *is surjective*,

(iii) R_0 *prorepresents the functor* $[\underline{A}]$ *over* $\underline{C}_\Lambda$.

Proof: We have already noted (a). As for (b), (i) $\Leftrightarrow$ (ii) is clear whereas (ii) $\Leftrightarrow$ (iii) follows from 2.7 and 2.15, since $\tilde{\xi} : h_{R_0} \longrightarrow [\underline{A}]$ is minimally smooth.

Remark 2.21. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$, and let $\underline{A}|\underline{C}_k$ be the pull-back of $p : \underline{A} \longrightarrow \underline{C}_\Lambda$ under $\underline{C}_k \hookrightarrow \underline{C}_\Lambda$. Then it is trivial to see that if $\underline{A}$ is prorepresented by $(R_0, R_1, \ldots)$ via (ξ, φ), then $\underline{A}|\underline{C}_k$ is prorepresented by $(k \hat{\otimes}_\Lambda R_0, k \hat{\otimes}_\Lambda R_1, \ldots)$ via $(k \hat{\otimes}_\Lambda \xi, k \hat{\otimes}_\Lambda \varphi)$ where $k \hat{\otimes}_\Lambda \xi$ = its direct image of ξ under the map $R_0 \longrightarrow k \hat{\otimes}_\Lambda R_0$, $k \hat{\otimes}_\Lambda \varphi$ = the image of φ under $\mathrm{Isom}_{\underline{A}(R_0)}(\xi(s_1), \xi(s_2)) \longrightarrow \mathrm{Isom}_{\underline{A}(k \hat{\otimes}_\Lambda R_1)}(k \hat{\otimes}_\Lambda \xi(s_1), k \hat{\otimes}_\Lambda \xi(s_2))$. If $(R_0, R_1, \ldots)$ is normalized, then so is $(k \hat{\otimes}_\Lambda R_0, k \hat{\otimes}_\Lambda R_1, \ldots)$, and the relative dimension of R_1 over R_0 is equal to the relative dimension of $k \hat{\otimes}_\Lambda R_1$ over $k \hat{\otimes}_\Lambda R_0$ since R_1 is smooth over R_0.

Remark 2.22. Let $\underline{A}$ be prorepresented by smooth cogroupoid $(R_0, R_1, \ldots)$ and $(R_0', R_1', \ldots)$ where we assume that $(R_0, R_1, \ldots)$ is normalized. We then obtain a homomorphism $(i_0, i_1) : (R_0, R_1, \ldots) \longrightarrow (R_0', R_1', \ldots)$. Now R_0' is a formal power-series ring over R_0 by 2.10(b), and the homomorphism (i_0, i_1) induces an isomorphism $R_0' \hat{\otimes}_{R_0} (R_0, R_1, \ldots) \xrightarrow{\sim} (R_0', R_1', \ldots)$.

3. The coherent sheaves $\underline{D}_X^i$

One of the nice functors which is usually not pro-representable but nevertheless admits a formal versal object is the deformation functor. To this end we recall a few basic facts on commutative ring extensions and its obstruction theory. This section contains nothing new (possibly except in its systematic approach via Grothendieck cohomology on site). A purpose of this section is to make this expose as much self-contained. (Indeed we prove here more than we need later). The basic facts are stated in theorem 3.12. Throughout rings are meant to be commutative rings with unit.

Given a ring R we denote by $\underline{C}_R$ the category of R-algebras. For each object A in $\underline{C}_R$ we set

$\mathrm{Cov}(A)$ = the set of all surjective arrows $\varphi : A' \to A$ in $\underline{C}_R$ with target A.

It clearly defines a pretopology and in turn we obtain a site $\underline{C}_R$. For each object A in $\underline{C}_R$, we shall denote by $\underline{C}_{A|R}$ the category of arrows in $\underline{C}_R$ with target A. We may and shall provide the category $\underline{C}_{A|R}$ with the topology induced from the forgetful functor $j_A : \underline{C}_{A|R} \to \underline{C}_R$. We note that the category $\underline{C}_{A|R}$ has a final object $(1_A : A \to A)$ and an initial object $(i_A : R \to A)$, and admits a fibre-product as well as coproduct. Henceforth we shall denote the object $(B \to A)$ in $\underline{C}_{A|R}$, by abuse of notation, simply by B in case when there is no possible confusion. Given an abelian category $\underline{C}$, a $\underline{C}$-valued sheaf on $\underline{C}_{A|R}$ is, by definition, a functor $F : \underline{C}^{o}_{A|R} \to \underline{C}$ such that, for any surjective arrow $B' \to B$ in $\underline{C}_{A|R}$,

$$0 \to F(B) \to F(B') \rightrightarrows F(B' \underset{B}{\times} B')$$

is exact. The <u>additive</u> presheaves are those which preserve coproduct i.e., $F(B_1 \underset{R}{\otimes} B_2) = F(B_1) \oplus F(B_2)$. For example, an A-module M defines the functor

$$\mathrm{Der}_R(-,M) : \underline{C}^{o}_{A|R} \to (A\text{-mod})$$

given by $B \to \mathrm{Der}_R(B,M)$ = R-linear derivations of B with values in M regarded as B-module via the structural map $B \to A$, and it is clearly an additive sheaf on $\underline{C}^{o}_{A|R}$, where (A-mod) denotes the category of A-modules. A presheaf F on $\underline{C}_{A|R}$ with values in (A-mod) is called "finite type" if $F(B)$ is an A-module of finite type whenever B is an R-algebra of finite type. For example, if A is noetherian and M is an A-module of finite type, then $\mathrm{Der}_R(-,M)$ is an additive sheaf of finite type on the site $\underline{C}_{A|R}$.

A few more words on notations and basic facts in homological algebra:

3.1(a) Throughout the rest of this section, we shall denote by $\underline{C}^{\sim}_{A|R}$ (or $\underline{C}^{\vee}_{A|R}$) the category of sheaves (or presheaves) with values in (A-mod). We have the (left exact) injection functor $i_{A|R} : \underline{C}^{\sim}_{A|R} \to \underline{C}^{\vee}_{A|R}$, and the associated-sheaf (exact) functor $G_{A|R}: \underline{C}^{\vee}_{A|R} \to \underline{C}^{\sim}_{A|R}$ which is adjoint to the functor $i_{A|R}$. An object $B \in ob(\underline{C}_{A|R})$ induces a continuous functor $j : \underline{C}_{B|R} \to \underline{C}_{A|R}$ and in turn a commutative diagram

$$(*) \qquad \begin{tikzcd} \underline{C}^{\sim}_{A|R} \arrow[r, "j^{\sim}"] \arrow[d, "i_{A|R}"'] & \underline{C}^{\sim}_{B|R} \arrow[d] \\ \underline{C}^{\vee}_{A|R} \arrow[r, "j^{\vee}"'] & \underline{C}^{\vee}_{B|R} \end{tikzcd}$$

where the restriction functor $\tilde{j}$ (as well as $j^{\vee}$) is an exact functor, since the site $\underline{C}_{B|R}$ is nothing but the category of arrows in $\underline{C}_{A|R}$ having target B , with the induced topology. A ring homomorphism $S \to R$ induces a continuous functor $\beta : \underline{C}_{A|R} \to \underline{C}_{A|S}$, and in turn a commutative diagram

$$(**) \qquad \begin{tikzcd} \underline{C}^{\sim}_{A|S} \arrow[r, "\tilde{\beta}"] \arrow[d, "i_{A|S}"'] & \underline{C}^{\sim}_{A|R} \arrow[d] \\ \underline{C}^{\vee}_{A|S} \arrow[r, "\beta^{\vee}"'] & \underline{C}^{\vee}_{A|R} \end{tikzcd}$$

Note that $\tilde{\beta}$ is not necessarily an exact functor.

3.1(b) The right derived functor of the injection functor $i_{A|R}: \underline{C}^{\sim}_{A|R} \to \underline{C}^{\vee}_{A|R}$ will be denoted by $\underline{H}^{\circ}_{A|R}$. For each sheaf F on $\underline{C}_{A|R}$, we set

$$H^{\circ}(A|R,F) = [H^{\circ}{}_{A|R}(F)](A)$$

For any $B' \to B$ in $Cov(B)$, let us define the functor $H^{\circ}((B' \to B),-) : \underline{C}^{\vee}_{A|R} \to$ (A-mod) by $H^{\circ}((B' \to B),F) = Ker(F(B') \rightrightarrows F(B' \times_B B'))$. The right derived functors of $H^{\circ}((B' \to B),-)$ will be denoted by $H^{\circ}((B' \to B),-)$. One knows that the $H^{\circ}((B' \to B),F)$ are the cohomology groups of the (Čech) semi-simplicial module

$$F(B') \rightrightarrows F(B' \times_B B') \rightrightarrows F(B' \times_B B' \times_B B') \rightrightarrows \cdots$$

The right derived functor of $\check{\underline{H}}^0_{A|R} : \underline{C}_{A|R} \to \underline{C}_{A|R}$ will be denoted by $\check{\underline{H}}^{\cdot}_{A|R}$,
where

$$[\check{\underline{H}}^0_{A|R}(F)](B) = \lim_{\{B' \to B\} \in \mathrm{Cov}(B)} H^0(\{B' \to B\}, F) \quad .$$

For each presheaf F on $\underline{C}_{A|R}$, we set

$$\check{H}^{\cdot}(A|R,F) = [\check{H}^{\cdot}_{A|R}(F)](A) = \lim_{\{A' \to A\} \in \mathrm{Cov}(A)} H^{\cdot}(\{A' \to A\}, F)$$

for any sheaf F on $\underline{C}_{A|R}$, we have the usual spectral sequence

$$\check{H}^p(A|R, \underline{H}^q_{A|R}(F)) \Rightarrow H^{p+q}(A|R,F)$$

where we note that $\check{H}^0(A|R,\underline{H}^q(F)) = 0$ for all $q > 0$.
In particular

$$\begin{cases} \check{H}^1(A|R,F) = H^1(A|R,F) \\ 0 \to \check{H}^2(A|R,F) \to H^2(A|R,F) \to \check{H}^1(A|R,\underline{H}^1_{A\ R}(F)) \to \check{H}^3(A|R,F) \quad \text{is exact.} \end{cases}$$

3.1(c) If B is an object in $\underline{C}_{A|R}$, the commutative diagram (*) in 2.1(a) together with the fact that $\tilde{j}$ as well as $j^{\vee}$ are exact functors yields the canonical isomorphism

$$H^{\cdot}(B|R,\tilde{j}F) = [\underline{H}^{\cdot}_{A|R}(F)](B)$$

On the other hand, a ring homomorphism $S \to R$ induces the commutative diagram (**) in 2.1(a). Now $\tilde{\beta}$ is not necessarily an exact functor. However, β preserves a fibre-product, and thus we obtain a spectral sequence

$$\underline{H}^p_{A|R}(R^q\tilde{\beta}F) \Rightarrow \check{\beta}\underline{H}^{p+q}_{A|S}(F) \qquad \text{i.e.,} \qquad H^p(A|R,R^q\tilde{\beta}F) \Rightarrow H^{p+q}(A|S,F)$$

Since we have the canonical isomorphism $\tilde{\beta} = \mathcal{G}_{A|R} \circ \check{\beta} \circ i_{A|S}$ and $\beta^{\#} = \mathcal{G}_{A|R} \circ \check{\beta}$ is an exact functor, we have $R^{\cdot}\widetilde{\beta}F = \beta^{\#}\underline{H}^{\cdot}_{A|S}(F)$. Thus the spectral sequence above can now be written as

$$H^p(A|R,\beta^{\#}\underline{H}^q_{A|S}(F)) \Rightarrow H^{p+q}(A|S,F) \quad .$$

Lemma 3.2. Given $B \in ob(\underline{C}_{A|R})$, choose a surjective arrow $P \to B$ in $\underline{C}_{B|R}$, where P is a polynomial algebra over R . Then for any $F \in ob(\underline{C}^{\vee}_{A|R})$, the natural map $H^{\cdot}(\{P \to B\},F) \to \check{\underline{H}}^{\cdot}(F)\ (B)$ is an isomorphism.

Proof: It suffice to see that the natural map

$$H^0(\{P \to B\},F) \to \check{\underline{H}}^0(F)\ (B) = \varinjlim_{\{B' \to B\} \in Cov(B)} H^0(\{B' \to B\},F)$$

is an isomorphism, which is clear since, for any $\{B' \to B\} \in Cov(B)$, there exists an arrow $\{P \to B\} \to \{B' \to B\}$.

Corollary 3.3.

(a) Assume that A is noetherian. If $F \in ob(\underline{\widetilde{C}}_{A|R})$ is of finite type, then $\underline{H}^n(F)$ is of finite type for all n .

(b) For any polynomial algebra P over R , we have $H^n(P|R,F) = 0$ for all $n > 0$ and for all $F \in ob(\underline{\widetilde{C}}_{P|R})$.

(c) A diagram $0 \to F' \to F \to F'' \to 0$ in $\widetilde{\underline{C}}_{A|R}$ is exact if and only if $0 \to F'(P) \to F(P) \to F''(P) \to 0$ is exact for every $P \in ob(\underline{C}_{A|R})$ which is a polynomial algebra over R .

Proof:

(a) In view of the spectral sequence

$$\check{H}^p(B|R,\underline{H}^q(F)) \Rightarrow H^{p+q}(B|R,F)$$

together with the fact that $\check{H}^0(\underline{H}^q(F)) = 0$ for all $q > 0$, it suffices to show that $\underline{H}^n(G)$ is of finite type for all n and for all presheaf G of finite type. However $\check{H}^{\cdot}(B|R,G) = H^{\circ}(\{P \to B\},G)$, where P is a polynomial algebra over R and $\{P \to B\} \in Cov(B)$, and furthermore $H^{\cdot}(\{P \to B\},G)$ is simply the cohomology gotten from the complex

$$\ldots\ldots\ldots\ldots\ldots G(P \underset{B}{\times} P \underset{B}{\times} P) \,\substack{\leftarrow\\\leftarrow\\\leftarrow}\, G(P \underset{B}{\times} P) \,\substack{\leftarrow\\\leftarrow}\, G(P) \; .$$

If B is an R-algebra of finite type, then we may take P to be of finite type over R so that $P\underset{B}{\times} \ldots\ldots \underset{B}{\times}P$ are all R-algebras of finite type. If G is of finite type, then $G(P\underset{B}{\times} \ldots\ldots \underset{B}{\times}P)$ are all A-modules of finite type, so that $H^{\cdot}(\{P \to B\},G)$ is an A-module of finite type since A is noetherian.

(b) Let P be a polynomial algebra over R . If $B \to P$ is any surjective R-algebra map, then it amits a section and hence $H^{i}(\{B \to P\},G) = 0$ for all $i > 0$, so that $\check{H}^{i}(P|R,\ G) = 0$ for all $i > 0$ and for all presheaves G on $\underline{C}_{P|R}$. It follows that the spectral sequence $\check{H}^{p}(P|R,\ \underline{H}^{q}_{P|R}(F)) \Rightarrow H^{p+q}(P|R,\ F)$ degenerates and hence $\check{H}^{0}(P|R,\ \underline{H}^{n}(F)) \simeq H^{n}(P|R,\ F)$ for all n . However $\check{H}^{0}(P|R,\ \underline{H}^{n}(F)) = 0$ for all $n > 0$ whenever F is a sheaf, and hence $H^{n}(P|R,\ F) = 0$ for all $n > 0$ and for all $F \in ob(\widetilde{\underline{C}}_{P|R})$.

(c) If $0 \to F' \to F \to F'' \to 0$ is an exact sequence in $\widetilde{\underline{C}}_{A|R}$, then

$$0 \to F'(B) \to F(B) \to F''(B) \to H'(B|R,\ F') \to H'(B|R,\ F) \to \ldots\ldots$$

is exact. If $P \in ob(\underline{C}_{A|R})$ is a polynomial algebra over R , then $H^{1}(P|R,\ F') = 0$ by 3.3(b) and hence

$$0 \to F' \to F \to F'' \to 0$$

is exact. As for the other direction, let $\underline{P}_{A|R}$ be the full subcategory of $\underline{C}_{A|R}$ in which

$$ob(\underline{P}_{A|R}) = \{P \in ob(\underline{C}_{A|R}) \mid \ P \text{ is a polynomial algebra over } R\} \ ,$$

provided with the induced topology, and let $p : \underline{P}_{A|R} \to \underline{C}_{A|R}$ be the inclusion functor. Since every object in $\underline{C}_{A|R}$ can be covered by an object in $\underline{P}_{A|R}$, it follows from a comparison lemma (   ) that $\widetilde{p} : \widetilde{\underline{C}}_{A|R} \to \widetilde{\underline{P}}_{A|R}$ is an equivalence of categories. Consequently, if $0 \to F'(P) \to F(P) \to F''(P) \to 0$ is exact for every $P \in ob(\underline{P}_{A|R})$, then $0 \to \widetilde{p}F' \to \widetilde{p}F \to \widetilde{p}F'' \to 0$ is exact in $\widetilde{\underline{P}}_{A|R}$ and hence $0 \to F' \to F \to F'' \to 0$ is exact in $\widetilde{\underline{C}}_{A|R}$.

Remark 3.4. Every surjective arrow in $\underline{P}_{A|R}$ admits a section. It follows immediately that every presheaf on $\underline{P}_{A|R}$ is actually a sheaf i.e., $\underline{P}^{\sim}_{A|R} = \underline{P}_{A|R}$. Consequently we find that the restriction functor $\underline{C}^{\sim}_{A|R} \to \underline{P}^{\vee}_{A|R}$ is an equivalence of categories.

Lemma 3.5. Consider a diagram of R-algebras

$$\begin{tikzcd} & C \arrow[d] \\ B' \arrow[r, "\text{surjective}"'] & B \end{tikzcd}$$

and let S be an R-algebra.

(i) If $\mathrm{Tor}_1^R(B,S) = 0$, then the canonical map

$$(B' \underset{B}{\times} C) \underset{R}{\otimes} S \to (B' \underset{R}{\otimes} S) \underset{(B \underset{R}{\otimes} S)}{\times} (C \underset{R}{\otimes} S)$$

is an isomorphism.

(ii) Assume that B' is R-flat. If $\mathrm{Tor}_i^R(C,S) = 0$ for all $0 < i < n$ and $\mathrm{Tor}_i^R(B,S) = 0$ for all $0 < i < n$, then $\mathrm{Tor}_i^R(B' \underset{B}{\times} C,\ S) = 0$ for all $0 < i < n$.

Proof: Let $0 \to J \to B' \to B \to 0$ be exact.

(i) If $\mathrm{Tor}_i^R(B,\ S) = 0$, then

$$0 \to J \underset{R}{\otimes} S \to B' \underset{R}{\otimes} S \to B \underset{R}{\otimes} S \to 0$$

is exact so that

$$0 \to J \underset{R}{\otimes} S \to (B' \underset{R}{\otimes} S) \underset{(B \underset{R}{\otimes} S)}{\times} (C \underset{R}{\otimes} S) \to C \underset{R}{\otimes} S \to 0$$

is exact. We thus obtain an exact commutative diagram

$$\begin{tikzcd} & J \underset{R}{\otimes} S \arrow[r] \arrow[d, equal] & (B' \underset{B}{\times} C) \underset{R}{\otimes} S \arrow[r] \arrow[d] & C \underset{R}{\otimes} S \arrow[r] \arrow[d, equal] & 0 \\ 0 \arrow[r] & J \underset{R}{\otimes} S \arrow[r] & (B' \underset{R}{\otimes} S) \underset{(B \underset{R}{\otimes} S)}{\times} (C \underset{R}{\otimes} S) \arrow[r] & C \underset{R}{\otimes} S \arrow[r] & 0 \end{tikzcd}$$

and therefore $(B'\otimes_B C)\otimes_R S \to (B'\otimes_R S) \times_{(B\otimes_R S)} (C \otimes_R S)$ is an isomorphism.

(ii) Since B' is R-flat and $\mathrm{Tor}_i^R(B, S) = 0$ for all $0 < i \le n$ by hypothesis, we have $\mathrm{Tor}_i^R(J, S) = 0$ for all $0 < i < n$. Then the exact sequence $0 \to J \to B'\times_B C \to C \to 0$ together with the hypothesis that $\mathrm{Tor}_i^R(C, S) = 0$ for all $0 < i < n$ entails that $\mathrm{Tor}_i^R(B'\times_B C, S) = 0$ for all $0 < i < n$.

Corollary 3.6.

(a) *Let A, S be R-algebras. If $\mathrm{Tor}_1^R(A, S) = 0$, then the canonical map $\check{H}^\cdot(A\otimes_R S|S, G) \to \check{H}^\cdot(A|R, SG)$ is an isomorphism for any $G \in \mathrm{ob}(\check{\underline{C}}_{A\otimes_R S|S})$ where the functor $S : \underline{C}_{A|R} \to \underline{C}_{A\otimes_R S|S}$ is given by $X \longmapsto X\otimes_R S$. In particular if S is R-flat, then the canonical map $\check{S}\check{H}^\cdot(G) \to \check{H}(\check{S}G)$ is an isomorphism for every $G \in \mathrm{ob}(\check{\underline{C}}_{A\otimes_R S|S})$.*

(b) *Let $P \to B$ be a surjective R-algebra map, where P is a polynomial algebra over R. If $\mathrm{Tor}_i^R(B, S) = 0$ for all $0 < i \le n$, then $\mathrm{Tor}_i^R(P\times_B P\times_B \ldots \times_B P, S) = 0$ for all $0 < i < n$.*

Proof:

(a) Given $B \in \mathrm{ob}(\underline{C}_{A|R})$, choose $\{P \to B\} \in \mathrm{Cov}(B)$ where P is a polynomial algebra over R. If $\mathrm{Tor}_1^R(B, S) = 0$, the natural map $(\overbrace{P\times_B \ldots \times_B P}^{n})\otimes_R S \to (P\otimes_R S)\times_{(B\otimes_R S)} \ldots \times_{(B\otimes_R S)}(P\otimes_R S)$ is an isomorphism for all n by 3.5(i), and hence $\check{H}^\cdot(B|R, \check{S}G) = H^\cdot(\{P \to B\}, \check{S}G) = H^\cdot(\{P\otimes_R S \to B\otimes_R S\}, G) = \check{H}^\cdot(B\otimes_R S|S, G)$.
If S is R-flat, then $\mathrm{Tor}_1^R(X, S) = 0$ for every $X \in \mathrm{ob}(\underline{C}_{A|R})$ and hence $\check{H}^\cdot(X\otimes_R S|S, G) \to \check{H}^\cdot(X|R, \check{S}G)$ is an isomorphism for every $X \in \mathrm{ob}(\underline{C}_{A|R})$ i.e. $\check{S}\underline{\check{H}}^\cdot(G) \to \underline{\check{H}}^\cdot(\check{S}G)$ is an isomorphism.

(b) Set $P^{(m)} = \overbrace{P\times_B \ldots \times_B P}^{m+1}$. If $m = 0$ there is nothing to prove since P is R-flat. Assume that $\mathrm{Tor}_i^R(P^{(m-1)}, S) = 0$ for all $0 < i < n$. Since $P \to B$ is surjective and $P^{(m)} = P\times_B P^{(m-1)}$, it follows from 3.5(ii) that $\mathrm{Tor}_i^R(P^{(m)}, S) = 0$ for all $0 < i < n$.

Proposition 3.7.

(a) Let A, S be R-algebras. If $\mathrm{Tor}_i^R(A, S) = 0$ for all $0 < i \leq n$, then for any sheaf F on $\underline{C}_{A \otimes_R S|S}$, the canonical map $H^i(A \otimes_R S|S, F) \to H^i(A|R, \tilde{S}F)$ is an isomorphism for all $i \leq n$, where $S : \underline{C}_{A|R} \to \underline{C}_{A \otimes_R S|S}$ is given by $X \longmapsto X \otimes_R S$.

(b) Let P be a polynomial algebra over R. Then for any additive sheaf F on $\underline{C}_{P \otimes_R A|R}$, the canonical map $H^n(P \otimes_R A|R, F) \to H^n(A|R, F)$ is an isomorphism for all $n > 0$.

(c) Let $S \to R \to A$ be ring homomorphisms. Then for any additive sheaf F on $\underline{C}_{A|S}$, we have an exact sequence

$$0 \to H^0(A|R, F_R) \to H^0(A|S, F) \to H^0(R|S, F) \to H^1(A|R, F_R) \to H^1(A|S, F) \to H^1(R|S, F) \to$$
$$\ldots\ldots\ldots \to H^n(A|R, F_R) \to H^n(A|S, F) \to H^n(R|S, F) \to H^{n+1}(A|R, F_R) \to \ldots.$$

where F_R is a sheaf on $\underline{C}_{A|R}$ defined by

$$F_R(X) = \mathrm{Ker}(F(X) \to F(R))$$

for all $X \in \mathrm{ob}(\underline{C}_{A|R})$ via the forgetful functor $\underline{C}_{A|R} \to \underline{C}_{A|S}$.

Proof:

(a) Consider the continuous functor

$$S : \underline{C}_{A|R} \to \underline{C}_{A \otimes_R S|S}$$

where $S(X) = X \otimes_R S$. Since the functor S sends a polynomial algebras over R to polynomial algebras over S, $\tilde{S} : \tilde{\underline{C}}_{A \otimes_R S|S} \to \tilde{\underline{C}}_{A|R}$ is an exact functor by 3.3(c) and in turn we obtain the canonical map $\check{S}\underline{H}^{\cdot}_{A \otimes_R S|S}(F) \to \underline{H}^{\cdot}_{A|R}(\tilde{S}F)$, and in turn we obtain the commutative diagram of spectral sequences:

$$\begin{array}{ccc}
E_S^{p,q}(S(X)) = \check{H}^p(X \otimes_R S|S, \underline{H}^q_{A \otimes_R S|S}(F)) & \Longrightarrow & H^{p+q}(X \otimes_R S|S, F) \\
\downarrow & & \\
\check{H}^p(X|R, \check{S}\underline{H}^q_{A \otimes_R S|S}(F)) & & \downarrow \\
\downarrow & & \\
E_R^{p,q}(X) = \check{H}^p(X|R, \underline{H}^q_{A|R}(\tilde{S}F)) & \Longrightarrow & H^{p+q}(X|R, \tilde{S}F)
\end{array}$$

We note that $E_S^{p,q}(S(X)) \to \check{H}^p(X|R, \check{S}\underline{H}^q_{A\otimes_R S|S}(F))$ is an isomorphism whenever $\mathrm{Tor}_1^R(X, S) = 0$ by 3.6(a). Assume now that $[\check{S}\underline{H}^q_{A\otimes_R S|S}(F)](X) \to [\underline{H}^q_{A|R}(\widetilde{S}F)](X)$ is an isomorphism for all $q < n$ and for all $X \in \mathrm{ob}(\underline{C}_{A|R})$ such that $\mathrm{Tor}_i^R(X, S) = 0$ for all 0 i n . It follows immediately from 3.6(b) that $E_S^{p,q}(S(X)) \to E_R^{p,q}(X)$ is an isomorphism for all $q < n$ and for all p , provided $\mathrm{Tor}_i^R(X, S) = 0$ for all $0 < i \leq n$. Since $E_S^{0,n}(S(X)) \to E_R^{0,n}(X)$ is an isomorphism (since both sides are zero), we get that $H^n(X_R S|S, F) \to H^n(X|R, \widetilde{S}F)$ is an isomorphism whenever $\mathrm{Tor}_i^R(X, S) = 0$ for all $0 < i \leq n$ i.e., $[\check{S}\underline{H}^i_{A\otimes_R S|S}(F)](X) \to [\underline{H}^i_{A|R}(\widetilde{S}F)](X)$ is an isomorphism for all $i \leq n$ whenever $\mathrm{Tor}_i^R(X, S) = 0$ for all $0 < i \leq n$.

(b) Let P be a polynomial algebra over R and let $f : \underline{C}_{A|R} \to \underline{C}_{A\otimes_R P|R}$ be the continuous functor given by $f(X) = X\otimes_R P$. Let $P' \to X$ be a surjective arrow in $\underline{C}_{A|R}$ where P' is a polynomial algebra over R . Then $P'\otimes_R P \to X\otimes_R P$ is a surjective arrow in $\underline{C}_{A\otimes_R P|R}$, and $P'\otimes_R P$ is again a polynomial algebra over R . Since P is flat over R , it follows from 3.5(i) that

$$\check{H}^{\cdot}(X|R, \check{f}G) = H^{\cdot}(\{P' \to X\}, \check{f}G) = H(\{P'\otimes_R P \to X\otimes_R P\}, G) = \check{H}^{\cdot}(X\otimes_R P|R, G)$$

for every $G \in \mathrm{ob}(\check{\underline{C}}_{A\otimes_R P|R})$. It follows that $\widetilde{f} : \widetilde{\underline{C}}_{A\otimes_R P|R} \to \underline{C}_{A|R}$ sends acyclic ones to acyclic ones. On the other hand, f sends polynomial algebras over R to polynomial algebras over R , and hence $\widetilde{f} : \widetilde{\underline{C}}_{A\otimes_R P|R} \to \widetilde{\underline{C}}_{A|R}$ is an exact functor by 3.2(c). It follows that $H^{\cdot}(A\otimes_R P|R, F) \to H^{\cdot}(A|R, \widetilde{f}F)$ is an isomorphism for every $F \in \mathrm{ob}(\widetilde{\underline{C}}_{A\otimes_R P|R})$. Now for each $X \in \mathrm{ob}(\underline{C}_{A|R})$, we have canonical maps

$$(\widetilde{f}F)\ (X) = F(X\otimes_R P) \to F(X)\otimes F(P) \quad ,$$

and hence a canonical map $\widetilde{f}F \to R\otimes F(P)$ where $F(P)$ denotes a constant sheaf (and F on the right hand side denotes, strictly speaking, the restriction of F on $\underline{C}_{A|R}$) . If F is an additive sheaf, then $(\widetilde{f}F)\ (X) = F(X\otimes_R P) \xrightarrow{\sim} F(X)\otimes F(P)$ is an isomorphism for every $X \in \mathrm{ob}(\underline{C}_{A|R})$ and hence $\widetilde{f}F \cong R\otimes F(P)$ is an isomorphism. Therefore

$$H^n(A|R, \tilde{f}F) \cong H^n(A|R, F\oplus F(P)) = H^n(A|R, F)$$

for all $n > 0$ since $F(P)$, being a constant sheaf, is flask. It follows that $H^n(A\otimes_R P|R, F) \to H^n(A|R, F)$ is an isomorphism for all $n > 0$, provided F is an additive sheaf on $\underline{C}_{A\otimes_R P|R}$.

(c) (communicated from Rinehart) Consider the forgetful functor $\beta : \underline{C}_{A|R} \to \underline{C}_{A|S}$. We have a spectral sequence

$$E^{p,q} = H^p(A|R, \beta^{\#}\underline{H}^q_{A|S}(F)) \Longrightarrow H^{p+q}(A|S, F)$$

where $\beta^{\#} = a_{A|R}\circ\check{\beta}$, $a_{A|R} : \check{\underline{C}}_{A|R} \to \tilde{\underline{C}}_{A|R}$ is "associated-sheaf" functor. Now let F be an additive sheaf. We claim that $\beta^{\#}\underline{H}^q_{A|S}(F)$ is a constant sheaf for all $q > 0$. Indeed, if P is a polynomial algebra over R, then $P = R\otimes_S P'$ for a polynomial algebra P' over S and hence

$$[\beta^{\#}\underline{H}^q_{A|S}(F)](P) = H^q(P|S, F) = H^q(R\otimes_S P'|S, F) \to H^q(R|S, F)$$

is an isomorphism for all $q > 0$ by 3.7(b), and surjective for $q = 0$. It follows from 3.3(c) that the canonical map $\beta^{\#}\underline{H}^q_{A|S}(F) \to H^q(R|S, F)$ = constant sheaf is an isomorphism for all $q > 0$, and is surjective for $q = 0$. It follows that $E^{p,q} = H^p(A|R, \beta^{\#}\underline{H}^q_{A|S}(F)) = 0$ for all $p, q > 0$, and hence the above spectral sequence yields the exact sequence

$$(*) \quad 0 \to E^{1,0} \to H^1 \to E^{0,1} \to E^{2,0} \to H^2 \to E^{0,2} \to E^{3,0} \to H^3 \to \ldots.$$

where we note that $E^{0,q} = [\beta^{\#}\underline{H}^q_{A|S}(F)](A) \cong H^q(R|S, F)$ for all $q > 0$ and $E^{p,0} = H^p(A|R, \tilde{\beta}F)$. On the other hand, the exact sequence $0 \to F_R \to \tilde{\beta}F \to F(R) \to 0$ gives us the exact sequence

$$\begin{array}{l} (**) \quad 0 \to H^0(A|R,F_R) \to H^0(A|R,\tilde{\beta}F) \to H^0(A|R,F(R)) \to H^1(A|R,F_R) \to H^1(A|R,\tilde{\beta}F) \to 0 \\ \qquad\qquad\qquad\qquad\qquad\qquad \| \qquad\qquad\qquad\qquad \| \\ \qquad\qquad\qquad\qquad\quad H^0(A|S,F) \qquad\quad H^0(R|S,F) \end{array}$$

The exact sequences (*) and (**) combined gives the desired exact sequence.

Corollary 3.8.

(a) If A, B are R-algebras such that $\operatorname{Tor}_i^R(A, B) = 0$ for all $0 < i \leq n$, then for any additive sheaf $F \in \operatorname{ob}(\widetilde{\underline{C}}_{A\otimes_R B|R})$, the canonical map

$$H^i(A\otimes_R B|R, F) \to H^i(A|R, F) \oplus H^i(B|R, F)$$

is an isomorphism for all $i \leq n$.

(b) Let A be an R-algebra such that $\operatorname{Tor}_i^R(A, A) = 0$ for all $i > 0$. If $A\otimes_R A \to A$ is an isomorphism, then for any additive sheaf $F \in \operatorname{ob}(\widetilde{\underline{C}}_{A|R})$, we have $H^i(A|R, F) = 0$ for all i.

Proof:

(a) Let $F \in \operatorname{ob}(\widetilde{\underline{C}}_{A\otimes_R B|R})$ be additive. By 3.7(c), the ring homomorphisms $R \to A \to A\otimes_R B$ gives us the exact sequence

$$(*) \quad \ldots \to H^i(A\otimes_R B\ A, F_A) \to H^i(A\otimes_R B\ R, F) \to H^i(A|R, F) \to \ldots.$$

consider now the commutative diagram

$$\begin{tikzcd}
H^i(A\otimes_R B|A, F_A) \arrow[r] \arrow[d] & H^i(A\otimes_R B|R, F) \arrow[d] \\
H^i(B|R, \widetilde{A}F_A) \arrow[r, "\approx"] & H^i(B|R, F)
\end{tikzcd}$$

where the functor $A : \underline{C}_{B|R} \to \underline{C}_{A\otimes_R B|A}$ is given by $X \longmapsto A\otimes_R X$, and the map $H^i(B|R, \widetilde{A}F_A) \to H^i(B|R, F)$ is induced from the map $\widetilde{A}F_A \to F$, which is an isomorphism since F is additive. Note that the commutativity of the above diagram, because of its functorial nature, is reduced to the case $i = 0$.

If $\operatorname{Tor}_i^R(A, B) = 0$ for all $0 < i \leq n$, then $H^i(A\otimes_R B|A, F_A) \to H^i(B|R, \widetilde{A}F_A)$ is an isomorphism for all $i \leq n$ by 3.7(a), and hence the composite map

$$H^i(A\otimes_R B|A, F_A) \to H^i(A\otimes_R B|R, F) \to H^i(B|R, F)$$

and (because of the symmetry)

$$H^i(A\underset{R}{\otimes}B|B,\ F_B) \to H^i(A\underset{R}{\otimes}B|R,\ F) \to H^i(A|R,\ F)$$

are isomorphisms for all $i \leq n$. This implies that the exact sequence (*) breaks up to short split-exact sequences

$$0 \to H^i(A\underset{R}{\otimes}B|A,\ F_A) \to H^i(A\underset{R}{\otimes}B|R,\ F) \to H^i(A|R,\ F) \to 0$$

for all $i \leq n$, and that

$$H^i(A\underset{R}{\otimes}B|R,\ F) \to H^i(A|R,\ F) \oplus H^i(B|R,\ F)$$

is an isomorphism for all $i \leq n$.

(b) Since $\mathrm{Tor}_i^R(A,A) = 0$ for all $i > 0$, it follows from 3.8(a) that

$$H^i(A\underset{R}{\otimes}A|R,\ F) \to H^i(A|R,\ F) \oplus H^i(A|R,\ F)$$

is an isomorphism for all i . On the other hand, $A\underset{R}{\otimes}A \to A$ is an isomorphism by hypothesis and hence $H^i(A|R,\ F) \to H^i(A\underset{R}{\otimes}A|R,\ F)$ is an isomorphism for all i . This means that the diagonal map $H^i(A|R,\ F) \to H^i(A|R,\ F) \oplus H^i(A|R,\ F)$ is an isomorphism for all i i.e., $H^i(A|R,\ F) = 0$ for all i .

Corollary 3.9: Let S be a multiplicative subset of an R-algebra A . If $F \in \mathrm{ob}\left(\underline{C}^{\sim}_{S^{-1}A|R}\right)$ is additive such that F_A is also additive, then the canonical map

$$H^i(S^{-1}A|R,\ F) \to H^i(A|R,\ F)$$

is an isomorphism for all i , where $F_A \in \mathrm{ob}\left(\underline{C}^{\sim}_{S^{-1}A|A}\right)$ is defined by $F_A(X) = \mathrm{Ker}(F(X) \to F(A))$ for all $X \in \mathrm{ob}\left(\underline{C}_{S^{-1}A|A}\right)$.

Proof: Since F is additive, the ring homomorphisms $R \to A \to S^{-1}A$ gives us the exact sequence

$$\cdots\cdots \to H^n(S^{-1}A|A,\ F_A) \to H^n(S^{-1}A|R,\ F) \to H^n(A|R,\ F) \to \cdots\cdots$$

On the other hand, since $S^{-1}A\underset{A}{\otimes}S^{-1}A \to S^{-1}A$ is an isomorphism and F_A is additive by hypothesis, it follows from 3.8(b) that $H^n(S^{-1}A|A,\ F_A) = 0$ for all n , and hence $H^n(S^{-1}A|R,\ F) \to H^n(A|R,\ F)$ is an isomorphism for all n .

Now let A be an R-algebra. An A-module M defines an additive sheaf $\mathrm{Der}_R(-, M) \in \mathrm{ob}(\tilde{\underline{C}}_{A|R})$ by $X \to \mathrm{Der}_R(X, M)$. We now make the following definition:

<u>Definition</u> 3.10: <u>Let</u> A <u>be an</u> R-<u>algebra. For each</u> A-<u>module</u> M <u>we set</u>

$$D^i(A|R, M) = H^i(A|R, \mathrm{Der}_R(-, M)) \quad .$$

We want to make an explicit computation of low-dimensional ones i.e., $D^i(A|R, M)$ for $i = 1, 2$: Choose a surjective arrow $P \to A$ in $\underline{C}_{A|R}$ where P is a polynomial algebra over R , and let $0 \to I \to P \to A \to 0$ be exact. Then

<u>Lemma</u> 3.10:

$$D^1(A|R, M) = \mathrm{Coker}(\mathrm{Der}_R(P, M) \to \mathrm{Hom}_P(I, M))$$
$$D^2(A|R, M) = H^1(A|R, \underline{H}^1_{A|R}(\mathrm{Der}_R(-, M)))$$

<u>Proof</u>: Consider the simplicial algebra

$$(*) \qquad \cdots\cdots\cdots \; P_A^{(2)} \mathrel{\substack{\to\\\to\\\to}} P_A^{(1)} \rightrightarrows P_A^{(0)}$$

where $P_A^{(n)} = \overbrace{P\underset{A}{\times}\cdots\underset{A}{\times}P}^{n+1}$. For each n we have the exact sequence

$$0 \to I^{(n)} \to P_A^{(n)} \to A \to 0$$

where $I^{(n)} = \overbrace{I\times I\times\cdots\times I}^{n+1}$, and it is easy to see that

(a):

$$0 \to \mathrm{Hom}_A(\bar{I}^{(n)}/\Delta(\bar{I}), M) \to \mathrm{Der}_R(P_A^{(n)}, M) \xrightarrow{\mathrm{Der}(\Delta,\, M)} \mathrm{Der}_R(P, M) \to 0$$

is exact for all n , where $\bar{I} = I/I^2$ and Δ stands for the diagonal map.
We also note the exact sequence

(b):

$$0 \to \mathrm{Hom}_A(\bar{I}^{(n)}/\Delta(\bar{I}), M) \to \mathrm{Hom}_A(\bar{I}^{(n)}, M) \xrightarrow{\mathrm{Hom}(\Delta,\, M)} \mathrm{Hom}_A(I, M) \to 0$$

The exact sequences (a) and (b) yield the respective exact sequences of simplicial A-modules. Since $\{I \to 0\}$, $\{I \to I\}$ admits sections, we have

$$H^i(\{I \to 0\}, \mathrm{Hom}_A(-, M)) = H^i(\{I \to I\}, \mathrm{Hom}_A(-, M)) = 0$$

for all $i > 0$, and it follows readily that

$$\begin{cases} 0 \to \mathrm{Der}_R(A, M) \to \mathrm{Der}_R(P, M) \to \mathrm{Hom}_A(\bar{I}, M) \to H^1(\{P \to A\}, \mathrm{Der}_R(-, M)) \to 0 \\ \qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad \text{is exact} \\ H^i(\{P \to A\}, \mathrm{Der}_R(-, M)) = 0 \quad \text{for all } i > 1 \end{cases}$$

It follows from 3.1(b) that

$$\begin{aligned} D^1(A|R, M) &= \check{H}^1(A|R, \mathrm{Der}_R(-, M)) = H^1(\{P \to A\}, \mathrm{Der}_R(-, M)) \\ &= \mathrm{Coker}(\mathrm{Der}_R(P, M) \to \mathrm{Hom}_A(\bar{I}, M)) \\ &= \mathrm{Coker}(\mathrm{Der}_R(P, M) \to \mathrm{Hom}_P(I, M)) \quad , \\ D^2(A|R, M) &= H^1(A|R, \underline{H}^1_{A|R}(\mathrm{Der}_R(-, M))) \quad . \end{aligned}$$

Now choose a free P-module F and a P-linear map $F \xrightarrow{\gamma} P$ such that $\gamma(F) = I$. Associated to P-linear map $\gamma : F \to P$, there is a Koszul complex, which is simply denoted, by abuse of notation, by K_I.

<u>Lemma</u> 3.11: $D^2(A|R, M) \simeq \mathrm{Coker}(H^1(K_I, M) \to \mathrm{Hom}_A(H_1(K^I), M))$.

<u>Proof:</u> We have

$$\begin{aligned} D^2(A|R, M) &= H^1(A|R, H^1_{A|R}(\mathrm{Der}_R(-, M))) \\ &= \mathrm{Ker}(H^1(P_A^{(1)}|R, M) \rightrightarrows H^1(P_A^{(2)}|R, M)) \end{aligned}$$

since $[\underline{H}^1_{A|R}(\mathrm{Der}_R(-, M))](P_A^0) = [\underline{H}^1_{A|R}(\mathrm{Der}_R(-, M))](P) = 0$.

Let $0 \to K \to F \xrightarrow{\gamma} I \to 0$ be exact. If we set $P[F]$ to be the symmetric algebra over P generated by F, then $P[F]$ and $P[F]\otimes P[F]$ are polynomial algebras over R and we obtain the surjective arrows

$$P[F] \to P_A^{(1)} = P\times_A P \qquad \text{given by} \qquad X \to (0, \gamma(X))$$

$$P[F]\otimes_P P[F] \to P_A^{(2)} = P\times_A P\times_A P \qquad X\otimes 1 \to (0, \gamma(X), 0),\ 1\otimes X \to (0, 0, \gamma(X))$$

where $X \in F$. If we set $J = \mathrm{Ker}(P[F] \to P_A^{(1)})$ and $J' = \mathrm{Ker}(P[F]\otimes_P P[F] \to P_A^{(2)})$, then

$$J = (F) \cap (F_\gamma) = (K) + (F\cdot F_\gamma)$$
$$J' = (F\otimes 1 + 1\otimes F) \cap (F_\gamma\otimes 1 + 1\otimes F) \cap (F\otimes 1 + 1\otimes F_\gamma)$$
$$= (K\otimes 1 + 1\otimes K) + (F\cdot F_\gamma\otimes 1 + 1\otimes F\cdot F_\gamma) + (F\otimes F)$$

where F_γ = the ideal generated by $\{X - \gamma(X) | X \in F\}$. Now $\pi_i : P_A^{(2)} \to P_A^{(1)}$ $i = 0, 1, 2$ can be lifted to $\hat{\pi}_i : P[F]\otimes_P P[F] \to P[F]$ where, for each $X \in F$,

$$\pi_0 = \begin{cases} X\otimes 1 \to \gamma(X) - X \\ 1\otimes X \to X \end{cases} \qquad \pi_1 = \begin{cases} X\otimes 1 \to 0 \\ 1\otimes X \to X \end{cases} \qquad \pi_2 = \begin{cases} X\otimes 1 \to X \\ 1\otimes X \to 0 \end{cases} .$$

A trivial computation shows that $\pi_0 - \pi_1 + \pi_2$ sends $K\otimes 1 + 1\otimes K + 1\otimes F\cdot F_\gamma$ to zero, whereas it sends $(F\cdot F_\gamma\otimes 1 + F\otimes F)$ onto $(F\cdot F_\gamma)$. Now for any $D \in \mathrm{Der}_R(P[F]\otimes_P P[F], M)$, $D(F\cdot F_\gamma\otimes 1 + F\otimes F) = 0$ since $I\cdot M = 0$, and consequently

$$\begin{aligned} D^2(A|R, M) &= \mathrm{Ker}(D^1(P_A^{(1)}|R, M) \rightrightarrows D^1(P_A^{(2)}|R, M)) \\ &= \{[f] \in D^1(P_A^{(1)}|R, M) | f(F\cdot F_\gamma) = 0\} \\ &= \mathrm{Coker}(\mathrm{Der}_R(P[F], M) \to \mathrm{Hom}_{P[F]}((F) \cap (F_\gamma)/(F\cdot F_\gamma), M)) \\ &= \mathrm{Coker}(\mathrm{Der}_P(P[F], M) \to \mathrm{Hom}_{P[F]}((F) \cap (F_\gamma)/(F)\cdot(F_\gamma), M)) . \end{aligned}$$

Now
$$0 \to (F) \to P[F] \xrightarrow{P[0]} P \to 0$$
$$0 \to (F_\gamma) \to P[F] \xrightarrow{P[\gamma]} P \to 0 .$$

are exact and hence $(F) \cap (F_\gamma)/(F)\cdot(F_\gamma) = \mathrm{Tor}_1^{P[F]}(P, P_\gamma) = H_1(K_I)$ where K_I is the Koszul complex associated to the P-linear map $F \xrightarrow{\gamma} I(\subset P)$. We thus have

$$\begin{aligned} D^2(A|R, M) &= \mathrm{Coker}(\mathrm{Hom}_P(F, M) \to \mathrm{Hom}_P(H_1(K_I), M)) \\ &= \mathrm{Coker}(H^1(K_I, M) \to \mathrm{Hom}_A(H^1(K_I), M)) . \end{aligned}$$

We can now state all the basic facts needed for our purpose, summarizing the various results we have already established.

oc

Theorem 3.12.

(1) *If* $0 \to M' \to M \to M'' \to 0$ *is an exact sequence of* A-*modules, then*

$$0 \to D^0(A|R, M') \to D^0(A|R, M) \to D^0(A|R, M'') \to D^1(A|R, M') \to$$
$$D^1(A|R, M) \to D^1(A|R, M'') \to \ldots\ldots\ldots \to D^n(A|R, M') \to$$
$$D^n(A|R, M) \to D^n(A|R, M'') \to \ldots$$

is exact.

(2) *Let* $S \to R \to A$ *be ring homomorphisms. Then for any* A-*module* M *we have the exact sequence*

$$0 \to D^0(A\ R, M) \to D^0(A\ S, M) \to D^0(R\ S, M) \to D^1(A\ R, M) \to$$
$$D^1(A\ S, M) \to D^1(R\ S, M) \to \ldots\ldots\ldots \to D^n(A\ R, M) \to$$
$$D^n(A\ S, M) \to D^n(R\ S, M) \to \ldots.$$

(3) *Let* A, S *be R-algebras such that* $\mathrm{Tor}_i^R(A, S) = 0$ *for all* $i > 0$. *Then for any* $A\otimes_R S$-*module* M , *the canonical map* $D^i(A\otimes_R S|S, M) \to D^i(A|R, M)$ *is an isomorphism for all* i . *In particular, if* A *is* R-*flat we have*

$$D^i(A\otimes S|S, M) \simeq D^i(A|R, M)$$

for all i *and for all* R-*algebras* S .

(4) *Let* S *be a multiplicative subset of* A . *Then for any* $S^{-1}A$-*module* N , *the canonical map*

$$D^i(S^{-1}A|R, N) \to D^i(A|R, N)$$

is an isomorphism for all i . *Consequently, for any* A-*module* M *we have the canonical isomorphism*

$$D^i(S^{-1}A|R, S^{-1}M) \simeq S^{-1}D^i(A|R, M)$$

for all i .

(5) *If* A, B *are* R-*algebras such that* $\mathrm{Tor}_i^R(A, B) = 0$ *for all* $0 < i \leq n$, *then for any* $A\otimes_R B$-*module* M , *the canonical map*

$$D^i(A\otimes_R B|R, M) \to D^i(A|R, M) \oplus D^i(B|R, M)$$

is an isomorphism for all $i \leq n$. *In particular, if* $\mathrm{Tor}_i^R(A, A) = 0$ *for all*

$i > 0$ *and*

$$\Delta : \operatorname{Spec}(A) \to \operatorname{Spec}(A) \underset{\operatorname{Spec}(R)}{\times} \operatorname{Spec}(A)$$

is an open immersion (for instance A/R *etale), then* $D^i(A|R, M) = 0$ *for all* $i > 0$ *and for all A-modules* M .

(6) *Let* A *be an R-algebra of essentially finite type over* R *, where* R *is noetherian, and let* E *be a flat A-module. Then we have a canonical isomorphism*

$$E \underset{A}{\otimes} D^i(A|R, M) \xrightarrow{\sim} D^i(A|R, E \underset{A}{\otimes} M)$$

for all $i \geq 0$ *and for all A-modules* M .

(7) *If* R *is noetherian and* A *is an R-algebra of essentially finite type, then* $D^i(A|R, M)$ *is A-module of finite type for all* i *, provided* M *is an A-module of finite type.*

(8) *Let* P *be a polynomial algebra over* R *and let* $0 \to I \to P \to A \to 0$ *be exact. Then* $D^0(A|R, M) = \operatorname{Der}_R(A, M)$,

$$D^1(A|R, M) = \operatorname{Coker}(\operatorname{Der}_R(P, M) \to \operatorname{Hom}_A(I/I^2, M))$$

= *the set of isomorphism classes of R-algebra extensions of* A *by* M ,

$$D^2(A|R, M) = \operatorname{Coker}(H^1(K_I, M) \to \operatorname{Hom}_A(H_1(K_I), M)) .$$

Proof:

(1) If $0 \to M' \to M \to M'' \to 0$ is an exact sequence of A-modules, then for any polynomial algebra P over R , $0 \to \operatorname{Der}_R(P, M') \to \operatorname{Der}_R(P, M) \to \operatorname{Der}_R(P, M'') \to 0$ is exact and hence $0 \to \operatorname{Der}_R(-, M') \to \operatorname{Der}_R(-, M) \to \operatorname{Der}_R(-, M'') \to 0$ is an exact sequence of sheaves by 3.3(c).

(2) $\operatorname{Der}_S(-, M)$ is an additive sheaf on $\underline{C}_{A|S}$, and $\operatorname{Der}_R(B, M) = \operatorname{Ker}(\operatorname{Der}_S(B, M) \to \operatorname{Der}_S(R, M))$ for every $B \in \operatorname{ob}(\underline{C}_{A|R})$. Therefore our assertion follows from 3.7(c).

(3) Let $S : \underline{C}_{A|R} \to \underline{C}_{A \otimes_R S|S}$ be the functor given by $X \longmapsto X \underset{R}{\otimes} S$. Then $\tilde{S}\operatorname{Der}_S(-, M) = \operatorname{Der}_R(-, M)$ and hence our assertion follows from 3.7(a).

(4) Follows from 3.9. Indeed, $F = \mathrm{Der}_R(-, N)$ is an additive sheaf on $\underline{C}_{S^{-1}A|R}$, and $F_A \in \mathrm{ob}\left(\underline{C}^{\sim}_{S^{-1}A|A}\right)$ is also additive since $F_A(X) = \mathrm{Ker}(F(X) \to F(A)) = \mathrm{Ker}(\mathrm{Der}_R(X, N) \to \mathrm{Der}_R(A, N)) = \mathrm{Der}_A(X, N)$ i.e., $F_A = \mathrm{Der}_A(-, N)$.

(5) The first assertion follows from 3.8(a). Now assume that $\mathrm{Tor}_i^R(A, A) = 0$ for all $i > 0$ and that

$$\Delta : \mathrm{Spec}(A) \to \mathrm{Spec}(A) \underset{\mathrm{Spec}(R)}{\times} \mathrm{Spec}(A)$$

is an open immersion i.e., $(A \otimes_R A)_{\varphi^{-1}(\underline{m})} \to A_{\underline{m}}$ is an isomorphism for all maximal ideals $\underline{m}$ of A, where $\varphi : A \otimes_R A \to A$ is the multiplication map. Then the first assertion combined with 3.12(4) entails that the diagonal map

$$\Delta_{\underline{m}} : D^i(A|R, M)_{\underline{m}} \to D^i(A|R, M)_{\underline{m}} \oplus D^i(A|R, M)_{\underline{m}}$$

is an isomorphism for every maximal ideal $\underline{m}$ of A and hence

$$\Delta : D^i(A|R, M) \to D^i(A|R, M) \oplus D^i(A|R, M)$$

is an isomorphism, which entails that $D^i(A|R, M) = 0$ for all i.

(6) In view of 3.12(4), we may assume that A is an R-algebra of finite type. Now consider the map

$$E \otimes_A \mathrm{Der}_R(-, M) \to \mathrm{Der}_R(-, E \otimes_A M)$$

in $\underline{C}^{\sim}_{A|R}$. Since E is A-flat, we have an isomorphism

$$E \otimes_A \mathrm{Der}_R(B, M) \xrightarrow{\sim} \mathrm{Der}_R(B, E \otimes_A M)$$

whenever $B \in \mathrm{ob}(\underline{C}_{A|R})$ is of finite type over R. It follows from 3.3(b) that

$$D^i(A|R, E \otimes_A \mathrm{Der}_R(-, M)) \to D^i(A|R, E \otimes_A M)$$

is an isomorphism. On the other hand, the functor $E \otimes_A - : (A\text{-mod}) \to (A\text{-mod})$ is exact, and therefore we have

$$E \otimes_A D^i(A|R, M) = D^i(A|R, E \otimes_A \mathrm{Der}_R(-, M)) ,$$

and hence our assertion.

(7) follows from 3.3(a).

(8) In view of 3.10 and 3.11, it suffices to show that $\mathrm{Coker}(\mathrm{Der}_R(P, M) \to \mathrm{Hom}_A(I/I^2, M)) = \mathrm{Exalcomm}(A|R, M)$, where $\mathrm{Exalcomm}(A|R, M)$ denotes the set of isomorphism classes of R-algebra extensions of A by M . Let $0 \to M \to A' \to A \to 0$ be an R-algebra extension of A by M . Since P is a polynomial algebra over R , we obtain a commutative diagram

$$\begin{tikzcd} 0 \arrow[r] & I \arrow[r] \arrow[d] & P \arrow[r] \arrow[d] & A \arrow[r] \arrow[d, equal] & 0 \\ 0 \arrow[r] & M \arrow[r] & A' \arrow[r] & A \arrow[r] & 0 \end{tikzcd}$$

where $I \to M$ is uniquely determined by the extensions A' up to $\mathrm{Der}_R(P, M)$. We thus obtain a map

$$\Phi : \mathrm{Exalcomm}(A|R, M) \to \mathrm{Coker}(\mathrm{Der}_R(P, M) \to \mathrm{Hom}_A(I/I^2, M)) \quad .$$

Conversely a P-linear map $I \to M$ yields an extension by push-out

$$\begin{tikzcd} 0 \arrow[r] & I \arrow[r] \arrow[d] & P \arrow[r] \arrow[d] & A \arrow[r] \arrow[d, equal] & 0 \\ 0 \arrow[r] & M \arrow[r] & A' \arrow[r] & A \arrow[r] & 0 \end{tikzcd}$$

Two P-linear maps $I \rightrightarrows M$ which differ by $\mathrm{Der}_R(P, M)$ is trivially seen to yield isomorphic extensions and therefore we obtain a map

$$\Psi : \mathrm{Coker}(\mathrm{Der}_R(P, M) \to \mathrm{Hom}_A(I/I^2, M)) \to \mathrm{Exalcomm}(A|R, M) \quad .$$

It is straightforward to verify that Φ, Ψ are inverse process to each other.

Corollary 3.13. (a) R-algebra A is formally smooth over R if and only if $D^1(A|R, M) = 0$ for every A-module M .

(b) Let R be noetherian and A an R-algebra of finite type. Then A is a relative complete-intersection over R if and only if $D^2(A|R, M) = 0$ for every A-module M .

(c) Let R be noetherian and A an R-algebra of finite type. Then for any multiplicative subset S in A , we have a canonical isomorphism $D^i(S^{-1}A|R, S^{-1}M) \xrightarrow{\sim} S^{-1}D^i(A|R, M)$ for all $i \geq 0$.

(d) Let R be noetherian and A a local R-algebra of essentially finite type. We denote by $\hat{A}$ the $\mathfrak{m}$-adic completion of A where $\mathfrak{m}$ = the maximal ideal of A. Then for any A-module E of finite type we have a canonical isomorphism

$$D^1(\hat{A}|R,\ \hat{A} \otimes_A E) \xrightarrow{\sim} \hat{A} \otimes_A D^1(A|R,\ E)\ .$$

Proof: (a) follows immediately from $D^1(A|R,\ M) = \mathrm{Exalcomm}(A|R,\ M)$.

(b) Let P be a polynomial algebra over R in a finite number of variables and let $0 \to I \to P \to A \to 0$ be exact. Localizing if necessary, we may assume that I is generated by a P-regular sequence which would imply that $H_1(K_I) = 0$ and therefore $D^2(A|R,\ M) = \mathrm{Coker}(H^1(K_I,\ M) \to \mathrm{Hom}_A(H_1(K_I),\ M)) = 0$. Conversely assume that $D^2(A|R,\ M) = 0$ for every A-module M.

Let $0 \to I \to P \to A \to 0$ be exact where P is a polynomial algebra over R in a finite number of variables. We have

$$0 = D^2(A|R,\ M) = \mathrm{Coker}(H^1(K_I,\ M) \to \mathrm{Hom}_A(H_1(K_I),\ M))$$

i.e., $H^1(K_I,\ M) \to \mathrm{Hom}_A(H_1(K_I),\ M)$ is surjective for all A-module M. Choose an exact sequence $0 \to K \to F \to I \to 0$ where F is a free P-module of finite type. We note that $H_1(K_I) = K/K_0$ where $K_0 = \mathrm{Im}(\overset{2}{\wedge} F \to \overset{1}{\wedge} F)$. The fact that $D^2(A|R,\ H_1(K_I)) = 0$ i.e.,

$$H^1(K_I,\ H_1(K_I)) \to \mathrm{Hom}_A(H_1(K_I),\ H_1(K_I)) \text{ is surjective}$$

entails that $0 \to K/K_0 \to F/IF \to I/I^2 \to 0$ is exact and splits as A-modules and in particular I/I^2 is projective as A-module. Therefore, localizing A if necessary, we may assume that I/I^2 is A-free, and that $F/IF \to I/I^2$ is an isomorphism. Then the fact that $0 \to K/K_0 \to F/IF \to I/I^2 \to 0$ is exact and splits as A-modules entails that $H_1(K_I) = K/K_0 = 0$. This means that I is generated by a P-regular sequence i.e., A is a relative complete intersection over R.

(c) follows immediately from 3.12(4) and (6).

(d) Firstly it follows from 3.12(6) that

$$D^i(A\ R,\ \hat{A} \otimes_A E) \xrightarrow{\sim} \hat{A} \otimes_A D^i(A|R,\ E)$$

is an isomorphism for all i, and therefore it suffices to show that

$D^1(\hat{A}|R, \hat{A}\otimes_A E) \to D^1(A|R, \hat{A}\otimes_A E)$ is an isomorphism. Our proof is based on the description of D^1 in terms of algebra extensions. Let $0 \to \hat{A}\otimes_A E \to A' \xrightarrow{\varphi} A \to 0$ be an R-algebra extension. The fact that $\hat{A}\otimes_A E$ is an $\hat{A}$-module of finite type entails readily that A' is a local R-algebra, complete with respect to $\underline{m}'$-adic topology where $\underline{m}'$ = the maximal ideal of A'. Therefore if there is an R-algebra map $\psi : A \to A'$ such that $\varphi \circ \psi = id_A$, then ψ extends uniquely to $\hat{\psi} : \hat{A} \to A'$ such that $\varphi \circ \hat{\psi} = id_{\hat{A}}$. This proves that

$$D^1(\hat{A}|R, \hat{A}\otimes_A E) \to D^1(A|R, \hat{A}\otimes_A E)$$

is injective. Now let $0 \to \hat{A}\otimes_A E \to B \xrightarrow{\varphi} A \to 0$ be an R-algebra extension. Firstly we claim that the topology on $\hat{E} = \hat{A}\otimes_A E$ induced from $\underline{m}_B$-adic topology on B coincides with the topology on $\hat{E}$ as $\hat{A}$-module. Indeed, since A is essentially of finite type over R, one can find a subalgebra B_0 of B such that B_0 is a local R-algebra of essentially finite type over R and that $\varphi(B_0) = A$. Set $E_0 = E \cap B_0$ and denote by $\underline{m}_B$, $\underline{m}_0$, $\underline{m}$ the maximal ideals of B, B_0, A respectively. Then $\underline{m}_B = \underline{m}_0 + \hat{E}$ and hence $\underline{m}_B^{n+1} = \underline{m}_0^{n+1} + \underline{m}^n\hat{E}$ for all n. Since B_0 is noetherian, there exists an integer $r > 0$ by Artin-Rees Theorem such that

$$\hat{E} \cap \underline{m}_B^{n+1} = \underline{m}^n\hat{E} + \hat{E} \cap \underline{m}_0^{n+1} = \underline{m}^n\hat{E} + E_0 \cap \underline{m}_0^{n+1}$$
$$= \underline{m}^n\hat{E} + \underline{m}^{n+1-r}(E_0 \cap \underline{m}_0^r) \subset \underline{m}^{n+1-r}\hat{E}$$

for all $n \geq r$, which proves our assertion about the two topologies on E. In particular $\hat{E} = \hat{A}\otimes_A E$ is complete with respect to the topology induced from $\underline{m}_B$-adic topology on B, and hence we obtain the exact commutative diagram

$$\begin{tikzcd}
0 \arrow[r] & \hat{E} \arrow[r] \arrow[d, equal] & B \arrow[r] \arrow[d] & A \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \hat{E} \arrow[r] & \hat{B} \arrow[r] & \hat{A} \arrow[r] & 0
\end{tikzcd}$$

where $\hat{B}$ stands for $\underline{m}_B$-adic completion of B. This proves that D^1
$D^1(\hat{A}|R, \hat{E}) \to D^1(A|R, \hat{E})$ is surjective and hence we are done.